SDC

Hackathon · Technical

Technical Tools Declaration

Lumina Nusantara · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

Submission deadline: 25 October 2026

> Technical Tools Declaration. Complete every section; where a category does not apply we ticked "No" or wrote "None" rather than leaving it blank.

1. TEAM AND PROJECT OVERVIEW

Team Name	Lumina Nusantara
Project / Solution Name	SkillRadar MY
Name (Leader)	DU YULIN
Email & Phone (Leader)	CYS2609009@xmu.edu.my / +60 11-1798 1756
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app
Code Repository URL (public)	https://github.com/douyinlin/sdc2026-lumina-nusantara

2. ARCHITECTURE & TECHNOLOGY STACK

2.1 System Architecture Overview

The Streamlit user interface is the only client judges and users touch. A user either uploads a synthetic CV or edits the questionnaire, both of which produce a skill profile normalised against a fixed vocabulary (feature #1). The UI reads a versioned trend snapshot from the data store; each record carries its source URL and retrieval date, and these are displayed alongside every forecast (feature #2). The matching and pathway engine consumes the profile plus the snapshot and returns ranked roles with a four-component explainable score, the exact skill gap (feature #3), an ordered reskilling plan (feature #4), SDG tags on every role (feature #5) and a moderate-versus-fast scenario comparison (feature #6). RunPod Serverless serves GPU inference for CV parsing and embeddings, with a local rule-based fallback so the demo degrades gracefully. A scheduled ingestion job refreshes trend snapshots from public sources. See the appended block diagram.

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	License
Python 3.11+	Application and engine language	PSF
Streamlit 1.66	Web UI, file upload, tables	Apache 2.0
pandas 2.x	Tabular handling	BSD 3-Clause
reportlab	Build these submission PDFs (offline tool)	BSD
sentence-transformers / all-MiniLM-L6-v2	Skill embeddings on the RunPod endpoint	Apache 2.0
unittest (stdlib)	Unit tests for the scoring engine	PSF

3. APIs & CLOUD SERVICES

3.1 External APIs

API and Provider	Purpose in Project	Access Tier
RunPod Serverless API — RunPod	GPU inference for CV parsing and embeddings	paid
WEF / O*NET / HRD Corp public data feeds	Trend snapshot ingestion (public data)	free
Streamlit Community Cloud	Hosting the public live demo	free
—	No other external API is called	—

No API keys or secrets are included in this document.

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Streamlit Community Cloud	Public live demo (auto-deploy from GitHub main)	Community / global
Replit	Development workspace and alternative hosting	Free tier
RunPod Serverless	GPU endpoint for CV parsing and embeddings	Community cloud
GitHub (douyinlin)	Source control and deployment trigger	Free

4. AI TOOLS, MODELS & DATASETS

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage	License / Terms
all-MiniLM-L6-v2	sentence-transformers (Hugging Face)	pre-trained	Apache 2.0
Optional hosted LLM for explanation text	Provider TBD	pre-trained	Provider terms
Rule-based skill extractor (in-house)	Team-built	self-trained (rules only)	Apache 2.0

4.2 Datasets

Dataset	Source / URL	Type	License	Personal data?
O*NET skills & occupations	onetcenter.org	public	CC BY 4.0	No
WEF Future of Jobs indicators	weforum.org	public	Cited	No
HRD Corp course directory	hrdcorp.gov.my	public	Public directory	No
Demo profiles & sample CV	Team-authored	synthetic	Apache 2.0	No

4.3 AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text, or other project material? **Yes**.

Tool & Version	Used For	Extent of Use
DeepSeek-based coding assistant	Scaffolding, boilerplate, documentation drafting	moderate
GitHub Copilot	Inline code completion	minor

Review statement: Every AI-generated block was read, tested or rewritten by a team member; matching logic, SDG tags and all statistics are hand-verified against the cited sources. Unit tests cover the scoring and pathway functions, and a reviewer other than the author checks each pull request before merge.

5. SPONSOR-PROVIDED RESOURCES

Did your project use any official SDC sponsor-provided resource? **Yes**.

Sponsor	Resource	How It Was Used in the Project
SDC 2026 programme	Replit workspace / cloud credits	Development environment and hosting of the prototype during the build phase
SDC 2026 programme	RunPod GPU credits	Serverless GPU inference for CV parsing and skill embeddings

Essentiality: The sponsor resources made GPU inference and zero-setup hosting possible within the hackathon window and let all five members work on the same running prototype. They are not permanently essential — the architecture keeps a local rule-based fallback for CV parsing and runs on any Streamlit host — so the solution survives if credits end.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1 Data Handling & Privacy

The public demo processes no real personal data. Users either pick a synthetic sample profile or upload a synthetic CV that the team authored; the interface states this rule next to the upload widget. Uploaded text is parsed in memory to extract skill names and is never written to disk, logged or stored in a database. No account, login or cookie is used beyond the Streamlit session itself. API keys live in platform secrets (Replit Secrets / Streamlit Cloud secrets) and never appear in the repository. If a future version accepts real CVs, it will require explicit consent, a retention limit and deletion on request.

6.2 Known Limitations & Risks

Main risks: (1) trend data lag and source bias — mitigated by storing source and retrieval date for every signal and showing them in the UI; (2) embedding mis-mapping of niche skills — mitigated by a curated alias vocabulary and a rule-based fallback; (3) hallucinated roles or courses — mitigated by keeping all roles and courses in curated tables with URLs and using any LLM only to phrase explanations; (4) dependency on a single GPU provider — mitigated by the local fallback; (5) small pilot size — statements are labelled as estimates. The tool supports human career decisions and never makes hiring decisions.

7. COMPLIANCE & READINESS

- ■ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ■ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ■ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ■ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ■ Any use of official SDC sponsor resources has been declared in Section 5.
- ■ The prototype described here is accessible to judges through the links provided in Section 1.
- ■ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: _____

Name: DU YULIN

Date: _____ / _____ / 2026